



COMcheck Software Version COMcheckWeb

Envelope Compliance Certificate

Project Information

Energy Code: 2020 New York City Energy Cons. Code, App. CA Modeling Envelope Backstop
Project Title: 79 Winthrop Street
Location: New York, New York
Climate Zone: 4a
Project Type: New Construction
Vertical Glazing / Wall Area: 20%
Performance Sim. Specs: EnergyPlus 8.1.0.009 (EPW: USA_NY_New.York-LaGuardia.AP.725030_TMY3.epw)

Construction Site: 79 Winthrop Street
Brooklyn, New York 11225
Owner/Agent:
Designer/Contractor:

Building Area

Floor Area

1-Multifamily : Residential	10059
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Envelope Assemblies

Assembly	Gross Area or Perimeter	Cavity R-Value	Cont. R-Value	Proposed U-Factor	Budget U- Factor _(a)
R1_Roof: Attic Roof, Steel Joists, [Bldg. Use 1 - Multifamily]	2327	26.4	15.0	0.027	0.020
FS1_Floor: Unheated Slab-On-Grade, Vertical 4 ft., [Bldg. Use 1 - Multifamily] (c)	209	---	8.0	0.504	0.520
NORTH					
W1_Ext. Wall: Concrete Block, 8in., Partially Grouted, Cells Empty, Light Density, Furring: Metal, [Bldg. Use 1 - Multifamily]	1779	13.0	8.0	0.063	0.086
D4_Door: Glass (over 50% glazing): Metal Frame, Non-Entrance Door, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	333	---	---	0.300	0.400
G2.1_Window: Metal Frame: Fixed, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	146	---	---	0.300	0.400
G1.1_Window: Metal Frame: Operable, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	519	---	---	0.300	0.400
W2_Ext. Wall: Steel-Framed, 16in. o.c., [Bldg. Use 1 - Multifamily]	264	22.0	8.0	0.056	0.061
D4_Door: Glass (over 50% glazing): Metal Frame, Non-Entrance Door, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	79	---	---	0.300	0.400
G4_Window: Metal Frame: Operable, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	12	---	---	0.300	0.400
F1_Ext. Wall: Solid Concrete, 12in. Thickness, Light Density, Furring: Metal, [Bldg. Use 1 - Multifamily]	353	13.0	8.0	0.060	0.086
G1.1_Window: Metal Frame: Operable, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	112	---	---	0.300	0.400
G2.1_Window: Metal Frame: Fixed, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	36	---	---	0.300	0.400

Assembly	Gross Area or Perimeter	Cavity R-Value	Cont. R-Value	Proposed U-Factor	Budget U- Factor ^(a)
D4_Door: Glass (over 50% glazing): Metal Frame, Non-Entrance Door, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	74	---	---	0.300	0.400
EAST					
W1_Ext. Wall: Concrete Block, 8in., Partially Grouted, Cells Empty, Light Density, Furring: Metal, [Bldg. Use 1 - Multifamily]	759	13.0	8.0	0.063	0.086
W2_Ext. Wall: Steel-Framed, 16in. o.c., [Bldg. Use 1 - Multifamily]	2806	22.0	8.0	0.056	0.061
SOUTH					
W1_Ext. Wall: Concrete Block, 8in., Partially Grouted, Cells Empty, Light Density, Furring: Metal, [Bldg. Use 1 - Multifamily]	1776	13.0	8.0	0.063	0.086
G1.1_Window: Metal Frame: Operable, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	590	---	---	0.300	0.400
D4_Door: Glass (over 50% glazing): Metal Frame, Non-Entrance Door, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	164	---	---	0.300	0.400
G2.1_Window: Metal Frame: Fixed, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	164	---	---	0.300	0.400
G3_Window: Metal Frame: Fixed, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	28	---	---	0.300	0.400
D1_Door: Glass (over 50% glazing): Metal Frame, Entrance Door, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	55	---	---	0.300	0.680
W2_Ext. Wall: Steel-Framed, 16in. o.c., [Bldg. Use 1 - Multifamily]	264	22.0	8.0	0.056	0.061
D2_Door: Insulated Metal, Swinging, [Bldg. Use 1 - Multifamily]	24	---	---	0.300	0.370
D8_Door: Glass (over 50% glazing): Metal Frame, Entrance Door, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	24	---	---	0.300	0.680
F1_Ext. Wall: Solid Concrete, 12in. Thickness, Light Density, Furring: Metal, [Bldg. Use 1 - Multifamily]	115	13.0	8.0	0.060	0.086
WEST					
W1_Ext. Wall: Concrete Block, 8in., Partially Grouted, Cells Empty, Light Density, Furring: Metal, [Bldg. Use 1 - Multifamily]	759	13.0	8.0	0.063	0.086
W2_Ext. Wall: Steel-Framed, 16in. o.c., [Bldg. Use 1 - Multifamily]	2802	22.0	8.0	0.056	0.061
G3_Window: Metal Frame: Fixed, Perf. Specs.: Product ID Alumtec, SHGC 0.30, VT 0.45, < 95' above-grade, [Bldg. Use 1 - Multifamily] (b)	22	---	---	0.300	0.400
D2_Door: Insulated Metal, Swinging, [Bldg. Use 1 - Multifamily]	24	---	---	0.300	0.370

(a) Budget U-factors are used for software baseline calculations ONLY, and are not code requirements.

(b) Fenestration product performance must be certified in accordance with NFRC and requires supporting documentation.

(c) Slab-On-Grade proposed and budget U-factors shown in table are F-factors.

Qualifies for 2020 NYCECC, App. CA Modeling : Envelope design 6% (allowable margin = -15.0%)

Envelope Compliance Statement

Compliance Statement: The proposed envelope design represented in this document is consistent with the building plans, specifications, and other calculations submitted with this permit application. The proposed envelope systems have been designed to meet the 2020 New York City Energy Cons. Code, App. CA Modeling Envelope Backstop requirements in COMcheck Version COMcheckWeb and to comply with any applicable mandatory requirements listed in the Inspection Checklist.

Name - Title	Signature	Date
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Inspection Checklist

Energy Code: 2020 New York City Energy Cons. Code, App. CA Modeling

Requirements: 0.0% were addressed directly in the COMcheck software

Text in the "Comments/Assumptions" column is provided by the user in the COMcheck Requirements screen. For each requirement, the user certifies that a code requirement will be met and how that is documented, or that an exception is being claimed. Where compliance is itemized in a separate table, a reference to that table is provided.

1	High Impact (Tier 1)	2	Medium Impact (Tier 2)	3	Low Impact (Tier 3)
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